1.1 Propositional Logic - 3

Friday, September 6, 2024 11:45 AM

Many of us are familiar with "boolean" variables for holding True vs. False.
We may do similar analysis using <u>bits</u>.

If x, y are boolean variables,

$x = 1 \Rightarrow$ <u>True</u>

$x = 0 \Rightarrow$ <u>False</u>

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	1
0	0	0	0	0

These operations can be extended to the case of longer strings of bits.

<u>Definition</u>: The <u>bitwise</u> $\wedge$, $\vee$, $\oplus$ is done by applying $\wedge$, $\vee$, $\oplus$ to the same digit placement of two bit strings.

$x = 101101 \qquad y = 011000$

```
  101101   x
  011000   y
  ------
  001000   bitwise ∧

  101101   x
  011000   y
  ------
  111101   bitwise ∨

  101101   x
  011000   y
  ------
  110101   bitwise ⊕
```

<u>Practice with translating</u>

Practice with translating

Let p = "You drove your car on the sidewalk."

q = "You got arrested."

Write in symbols...

- you do $\boxed{\text{not}}$ drive on the sidewalk. ($\sim p$)
- you will get arrested $\boxed{\text{if}}$ you drive on the sidewalk. ($p \Rightarrow q$)
- You get arrested $\boxed{\text{but}}$ you do not drive on the sidewalk. ($q \wedge \sim p$)

In general, the proposition following "if" is the hypothesis. Using words we can present the propositions in the reverse order of the implication.

The word "but" is odd. It should be understood in the above example that both propositions are occuring as in this <u>and</u> that. The "but" is just giving flavor and feels more natural when combined with a negation.

Write in english...

- $p \oplus q$
- $q \to \sim p$

- $p \Leftrightarrow q$

"I drive my car on the sidewalk if and only if I get arrested." This is explicit and acceptable.

$\Leftrightarrow$

"I only get arrested if I drive my car on the sidewalk." This is still acceptable but feels more natural.

- $p \oplus q$

"I drive my car on the sidewalk or I get arrested, but not both." explicit

"Either I drive my car on the sidewalk

or I get arrested, but not both.)

"Either I drive my car on the sidewalk or I get arrested." } more natural

- $q \Rightarrow \sim p$

"If I get arrested, I will not drive my car on the sidewalk."